

Class-2

3 bit Address bus and 4 bits Data Bus.

000	1011
001	0011
010	
011	
100	
101	1111
110	0000
111	

RAM

3 bit Address bus, so total 8 or (0-7)

4 bit Data bus, so 4 bit Data.
 Data.
 Data.
 Data.

NB if Address Bus $\rightarrow$ 20 bit

Address: 0 $\rightarrow (2^{20} - 1)$

Total Address: ~~20~~ 2^{20}

CPU is not RAM so, RAM is a external device.

Ex:

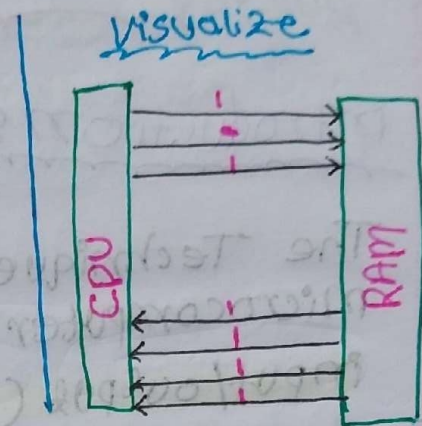
Read data from 5th index of RAM.

Step: 1

CPU → Address bus (101) → RAM

Step: 2

RAM → Data Bus (1111) → CPU



Basically mainly CPU index 5 is the address, it is the request made to Address Bus. RAM CPU is Data bus Data bus is the data.

Microcomputer input/output

introduction :

The Technique of data transfer between a microcomputer and an external device is called input/output (i/o)

Peripheral device:

External device যার মাধ্যমে motherboard এর সাথে computer এর সাথে connect করে দেওয়া হবে যার peripheral device.

input/output
↑

peripherals are the I/O devices that connected to a computer and communicate between the computer and the outside world.

interface:

এর মাধ্যমে (Logic + hardware) এর combination

We need interface hardware circuitry between the microcomputer and I/O devices.

Interface hardware provide all input and output transfers between the microcomputer and peripherals.

There are three ways of transferring data between a microcomputer and physical I/O devices.

I/O types

i) Programmed I/O:

I/O type:

I/O

i) programmed I/O:

In programmed i/o the data transfer is accomplished through an i/o port and controlled by software.

এখানে coding logic দিতে input/output device কে control করে।

এখানে আরো যেখানে জরুরি থাকবে বাক্য দি হবে।
CPU বাক্য ক্রম করে বা CPU দি throw হও এর হবে।

Ex:

```
int main()
{
    int a, b;
    a = 5;
    print(a)
    scan(b)
    print(b)
}
```

অর্থাৎ এখানে যেখানে জরুরি
আছে বাক্য input (scan) output
(print) হবে।

I/O port:

A microcomputer ~~communication~~ communicates with an external device via one or more registers called I/O ports using programmed I/O.

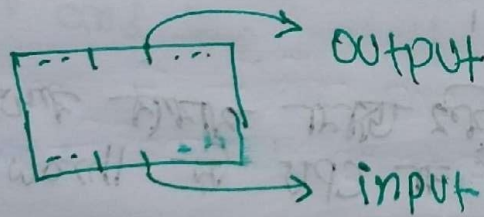
I/O ports are two types:

i) individual I/O:

Each ~~port~~ bit in the port can be configured individually as either input or output.

sets pin mode ~~individually~~ ~~trans~~ ~~output~~
pin ~~to~~ ~~first~~ ~~input~~ ~~output~~ ~~configure~~ ~~the~~ ~~as~~
~~the~~ ~~input/output~~ ~~pin~~.

Ex:



N.B

- i) I/O port is made up of group of 8 pins
- ii) Each pin can be configured as either input or output pin
- iii) input means: CPU $\leftarrow$ devices.
- iv) output means: CPU $\rightarrow$ devices.

Two types Register:

a) DDR

↳ Data direction Register

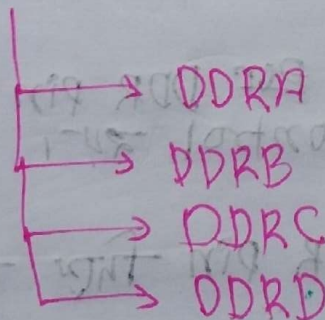
DDR Register for mainly port for pin for input and output to configured for 201.

N.B

i) if the bit of the DDR is 1 then the corresponding pin of the port is configured as output pin

ii) if the bit of DDR is 0 then the corresponding pin of the port is configured as input pin.

Types of DDR



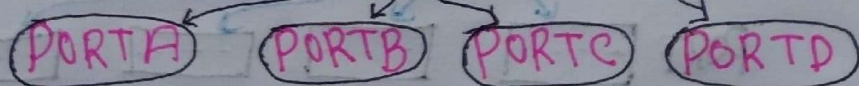
DDR

↳ 0: input (toward CPU)
↳ 1: output (toward I/O device)

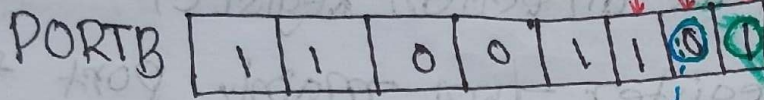
b) PORT Register

PORT Register original data carry/hold

PORT Register type



CPU



I/O device

DDR B जि चि बि bit जे zero आति जे error corresponding PORTB जि Data जे यह CPU जि मिले।

Individual & Gate per DDR pin in ~~gate~~
port & Gate pin control also.

parallel 4 बिटर DDR pin दिए बिटर PORT control डर।

N.B Working principle transistors

